Shiheng Zhang

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Seattle, WA
Google Scholar | GitHub

Research Interests

Numerical analysis, kinetic equations, dynamical low-rank approximation, structure-preserving computation, and scientific computing.

Education

Purdue University, West Lafayette, IN 09/2019 – 12/2023
Ph.D. in Computational and Applied Mathematics
Advisor: Jie Shen
Thesis: Energy-Dissipative Methods in Numerical Analysis, Optimization and Deep Neural Networks for Gradient Flows and Wasserstein Gradient Flows

Southern University of Science and Technology, Shenzhen, China 09/2015 – 06/2019
B.S. in Computational and Applied Mathematics
Advisors: Jiang Yang and Bingsheng He

Research Experience

University of Washington, Department of Applied Mathematics, Seattle, WA 01/2024 – Present
Postdoctoral Researcher with Jingwei Hu

- Dynamical low-rank algorithms for kinetic equations and related PDEs.
- Structure-preserving schemes and scientific machine learning for large-scale computation.

Awards and Honors

- Boeing Research Award, University of Washington, 2025
- T.T. Moh Graduate Scholarship, Purdue University, 2019
- Summa Cum Laude Graduate, Southern University of Science and Technology, 2019
- First Prize, Chinese Mathematics Competitions (Guangdong), Chinese Mathematical Society, 2017

Publications

Full publication list: Google Scholar

Journal Articles

J1. Einkemmer, L., Hu, J. & Zhang, S. Asymptotic-Preserving Dynamical Low-Rank Method for the Stiff Nonlinear Boltzmann Equation. *Journal of Computational Physics* (2025).

J2. Zhang, S. & Shen, J. Structure preserving schemes for a class of Wasserstein gradient flows. *Communications on Applied Mathematics and Computation.* doi:10.1007/s42967-025-00486-2 (2025).

J3. Zhang, S., Shen, J. & Hu, J. SAV-based entropy-dissipative schemes for a class of kinetic equations. *SIAM Journal on Scientific Computing* (2025).

J4. Zhang, S., Zhang, J., Shen, J. & Lin, G. A Relaxed Vector Auxiliary Variable Algorithm for Unconstrained Optimization Problems. *SIAM Journal on Scientific Computing.* doi:10.1137/23M1611087 (2025).

J5. Zhang, J., Zhang, S., Shen, J. & Lin, G. Energy-dissipative evolutionary deep operator neural networks. *Journal of Computational Physics* (2024).

Preprints

R1. Zhang, S. & Hu, J. *A Separable and Asymptotic-Preserving Dynamical Low-Rank Method for the Vlasov–Poisson–Fokker–Planck System* arXiv:2601.11900.

R2. Zhang, J., Zhang, S. & Lin, G. *Low-Rank Adaptation of Evolutionary Deep Neural Networks for Efficient Learning of Time-Dependent PDEs* arXiv:2509.16395.

R3. Zhang, S., Einkemmer, L. & Hu, J. *On the stability of the low-rank projector-splitting integrators for hyperbolic and parabolic equations* arXiv:2507.15192.

Talks and Posters

Talks

T1. Zhang, S. *On the stability of the low-rank projector-splitting integrator for parabolic equations* Talk, CHaRMNET Annual Meeting, University of Washington, December 2025. 2025.

T2. Zhang, S. *Structure preserving schemes for a class of Wasserstein gradient flows* Invited minisymposium talk, SIAM Annual Meeting, Spokane, WA, July 2024. 2024.

T3. Zhang, S. *A Relaxed Vector Auxiliary Variable Algorithm for Unconstrained Optimization Problems* Invited minisymposium talk, SIAM Conference on Optimization, Seattle, WA, June 2023. 2023.

Posters

P1. Zhang, S. *A Separable and Asymptotic-Preserving Dynamical Low-Rank Method for the Vlasov–Poisson–Fokker–Planck System* Poster, IPAM, UCLA, Los Angeles, CA, March 2026. 2026.

P2. Zhang, S. *SAV-based entropy-dissipative schemes for a class of kinetic equations* Poster, ICERM, Providence, RI, July 2025. 2025.

P3. Zhang, S. *Asymptotic-Preserving Dynamical Low-Rank Method for the Stiff Nonlinear Boltzmann Equation* Poster, CHaRMNET Annual Meeting, Virginia Tech, December 2024. 2024.

P4. Zhang, S. *SAV-based entropy-dissipative schemes for a class of kinetic equations* Poster, NSF Computational Mathematics PI Meeting, University of Washington, July 2024. 2024.

Teaching

Instructor, University of Washington

- AMATH 352 B: Applied Linear Algebra and Numerical Analysis – Winter 2026
- AMATH 351 A: Introduction to Differential Equations and Applications – Spring 2025
- AMATH 352 B: Applied Linear Algebra and Numerical Analysis – Winter 2025
- AMATH 351 A: Introduction to Differential Equations and Applications – Spring 2024

Teaching Assistant, Purdue University

- MA 265: Linear Algebra
- MA 266: Ordinary Differential Equations
- MA 261: Multivariate Calculus
- MA 162: Calculus II
- MA 161: Calculus I

Technical Skills

- **Programming:** Python (PyTorch, JAX, TensorFlow), Julia, C++, MATLAB, Git, SQL
- **Research Areas:** Numerical optimization, dynamical low-rank methods, scientific machine learning, numerical analysis for PDEs, structure-preserving schemes

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Last updated: May 3, 2026